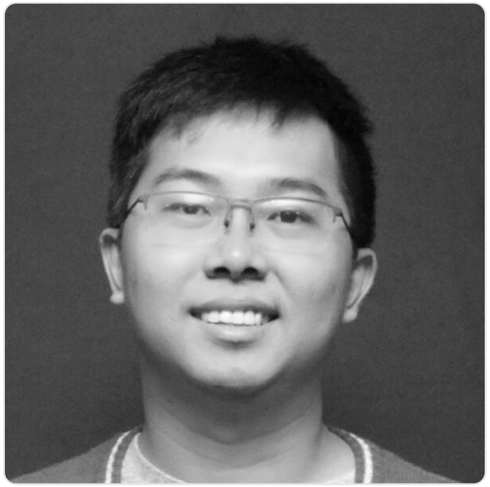
Five layers to tune before you fork your agent

The two cheapest layers, tools and skills, move quality the most.

Han Ngo

CTO, Dwarves



Han Ngo

Backend and AI systems engineer

CTO, Dwarves

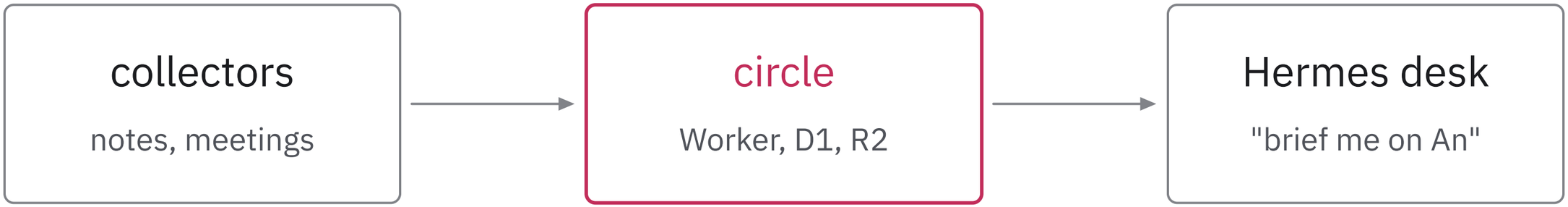
BUILT	Payment systems, a retail stock broker, a quant trading platform
NOW	LLM agents: memory, model routing, verification
ALSO	Contributes to Console Labs
MEMO	166 notes on memo.d.foundation

truonghan.com · memo.d.foundation/contributor/tieubao · github.com/tieubao · mail@truonghan.com

Better tool calls do not need a new agent.

The harness already runs the loop and manages the window.

circle stores people. Hermes talks to me.



So where is the agentic part?

Every step lands in the one window

1 ask: brief me on An

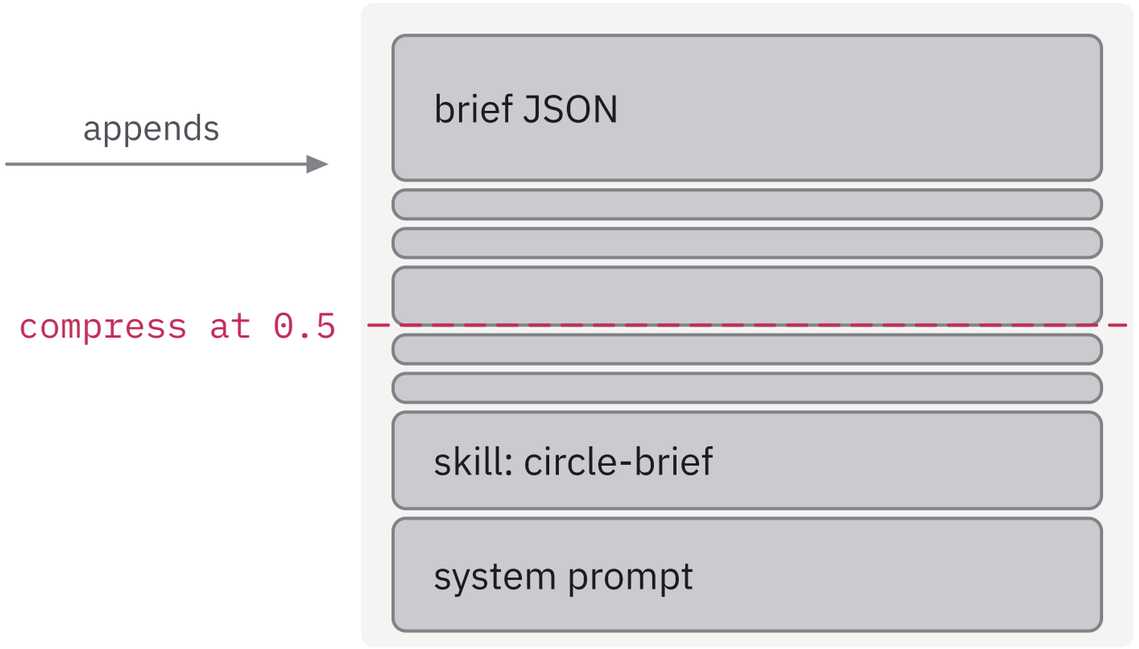
2 tool call

3 result: 2 candidates

4 back to me: which An?

5 tool call + result

6 done: one screen



context window: the model sees only this

Context management is a budget.

Tool lists, skills, results and old turns all spend from one fixed pot.

Agentic means the model picks the next step.

Your code or the model: that is the whole question.

Three kinds of work

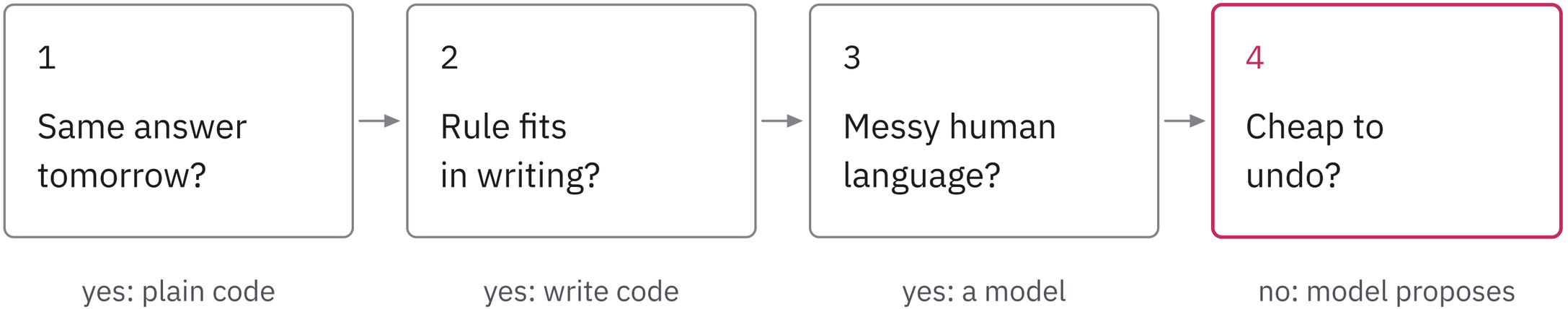
Plain code	Model on a fixed path	Agent
warmth score, radar	fact extraction	investigate, chat

the model takes control

Use the least agency that solves the task.

An agent costs tests, tokens and loop guards. Promote one only when the steps are unknown.

Four questions sort a feature



Ask them in order.

The seam is always a proposal

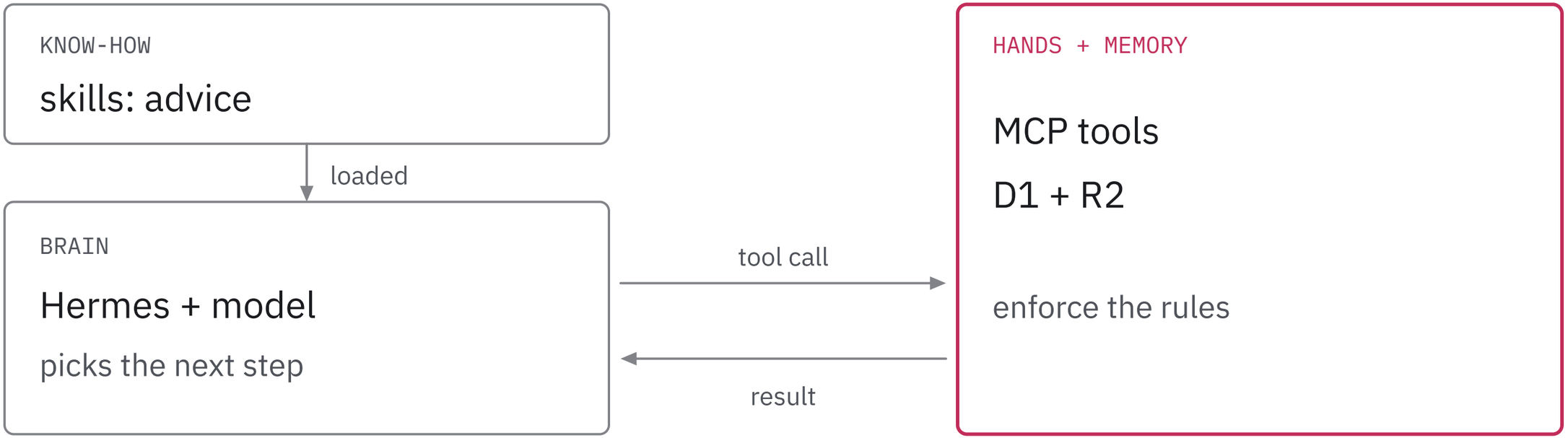


A person or a rule commits it.

circle has no agent loop of its own.

The loop belongs to whoever is calling.

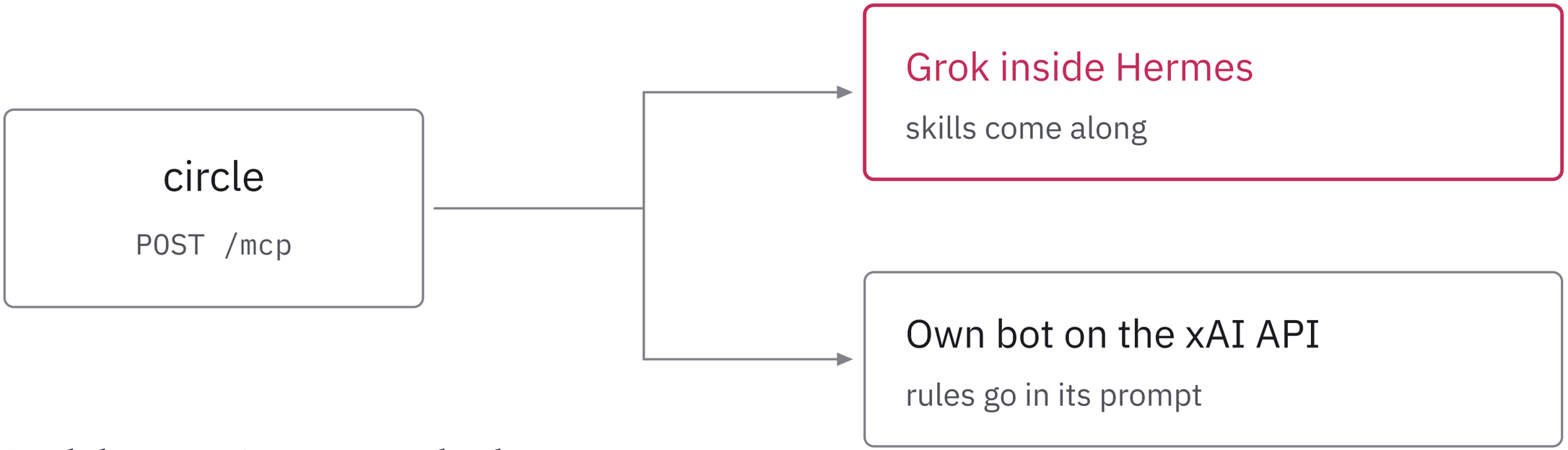
Brain, know-how, hands and memory



> Know-how is a suggestion.
> Hands are a guarantee.

The rule I keep from building circle

Grok gets the same hands two ways



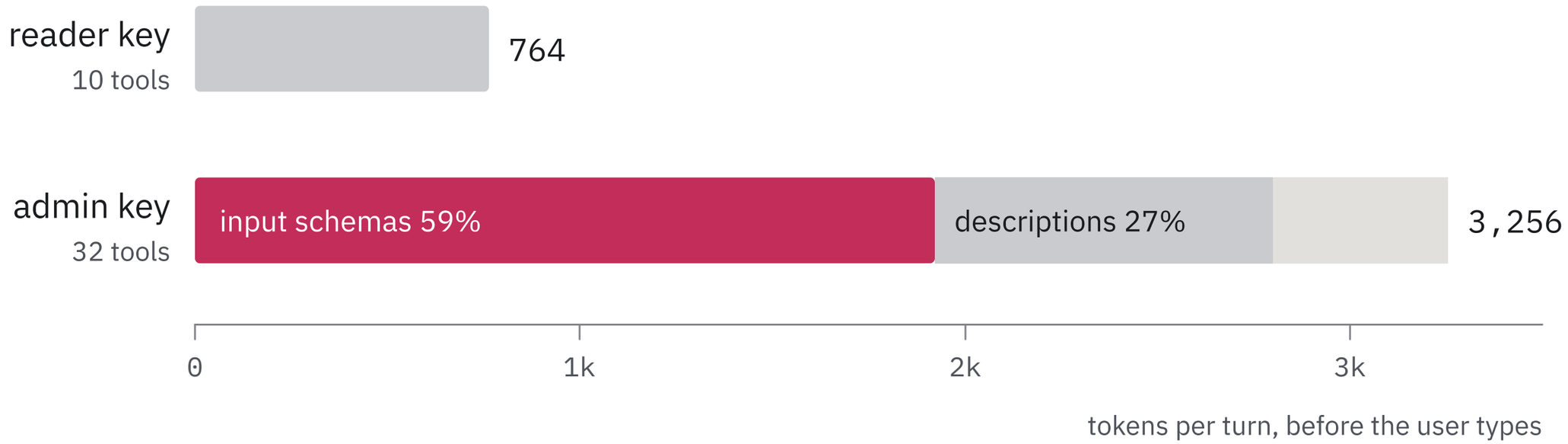
Each bot gets its own reader key.

Five layers, cheapest first

1	MCP tools	your code
2	Skills	your code
3	Harness config	config file
4	Engine patch	patch stack
5	Your own harness	to learn only

tune here first

Tool lists cost tokens every turn



Skills load on demand: 28,500 tokens become a 2,000-token index.

13 skills, one body in the window at a time.

A skill change reaches only new sessions.

17 sessions on 4 desks kept a retired tool and burned their budget.

The context dials are config

```
compression:
  enabled: true
  threshold: 0.5        # compact when the window is half full
  target_ratio: 0.2     # squeeze old turns to about a fifth
  protect_last_n: 20    # never compact the newest 20 messages
```

The loop-guard defaults live in the engine source.

Patch the engine only after layers 1 to 3 run out.

Write your own harness to learn. Keep it out of the serving path.

If a crash loses the work, it is a long chat.

A long-running agent keeps its state outside the window and reloads it.

Check these before you fork

- [ ] who decides?
- [ ] writes propose
- [ ] core is code
- [ ] fresh sessions
- [x] count tool tokens
- [ ] state outside window

Tune your tools and skills first. Fork the agent last.

Han Ngo

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